SI 315 - Models of Social Information Processing - Fall 2026

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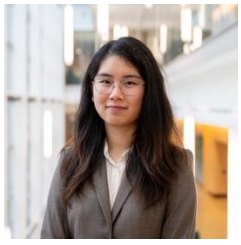


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Textbook
Title: “Networks, Crowds and Markets: Reasoning about a Highly Connected World”
Authors: David Easley and Jon Kleinberg
Access: You can freely download a pre-publication version of the book at the authors’ site: http://www.cs.cornell.edu/home/kleinber/networks-book/

Course Description
This course focuses on how social groups form, interact, and change. We look at the technical structures of social networks and explore how individual actions are combined to produce collective effects. This course introduces two conceptual models for how information flows is used in multi-person settings -- networks and games. Networks describe the structure of connections among people or documents. They permit mathematical analysis and meaningful visualizations that highlight different roles played by different people or documents, as well as features of the collection as a whole. Games describe the actions available to different people and how each person’s outcomes are contingent on the choices of others. They permit the analysis of stable sets of choices by all players (equilibrium) and provide a framework for the analysis of the theoretical effects of alternative designs for markets and information elicitation mechanisms. The techniques learned in this course can provide a high-level understanding of systems like Facebook, recommender systems used in sites such as Netflix, auction systems such as eBay, and information networks used by search engines such as Google.

Learning Objectives
- Identify different types of networks and describe a network using appropriate terminology such as node, edge, and connected component.
- Describe several theories of social and information network formation such as triadic closure, structural balance, and social influence.
- Given a small network, explain how much power each node has using terminology such as node centrality, outside option, and Nash bargaining solution.
- Describe the assumptions made by the HITS and Page Rank algorithms.
- Given a small network, find and interpret the PageRank, authority score, and hub score of each node.
- Describe a few different models of information cascades on social networks and analyze how a cascade will diffuse on small networks.
- Describe the Preferential Attachment and Small-World models.
- Analyze large networks using Python’s library NetworkX by applying concepts covered in the class.
- Identify the basic components of a game such as players, strategies, and payoffs.
- Analyze a game by finding and interpreting dominant strategies and pure and mixed strategy Nash equilibria.
- Identify several well-known games and real-world situations that can be modeled by these games.
- Identify different types of auctions and explain how each type will incentivize bidders to behave.

Tentative Weekly Topics

The schedule of topics and readings may change as the course progresses. Please check assignments and lecture slides on Canvas to see what has been covered so far.

Week of	Approximate Reading	Topics
Aug 31	Ch 1-2	Why study networks, Network connectivity, Distance, Breadth-first search, Small world experiment
Sept 7	Ch 3.1-3.5	Triadic closure, Clustering coefficient, Bridges, Strong and weak ties, Strength of weak ties.
Sept 14	Ch 4.1-4.4	Structural holes, Homophily, Selection and social influence, Affiliation networks.
Sept 21	Ch 5.1-5.4	Signed networks, Structural balance, Weak structural balance
Sept 28	Ch 6.1-6.3	Prisoners' Dilemma, Best response, Dominant Strategy
Oct 5	Ch 6.4-6.9 Ch 8.1-8.2	Finding mixed and pure strategy Nash equilibria, Coordination games. Network traffic, Braess' paradox,
Oct 12	Ch 9.1-9.5	Auctions **Exam 1 Review**
Oct 19 **Fall Break** **Exam 1**	Exam 1	Fall Break **Exam 1 on October 22, in class**
Oct 26	Ch10.1-10.5 Ch 12.1-12.3	Bipartite graphs, The Bipartite Matching Problem, Markets, Market clearing prices. Power in Networks, Network exchange experiments.
Nov 2	Ch 12.4-12.8 Ch 14.1-14.2	Stability and balance, The Ultimatum game. Link analysis and web search, Hubs and authorities, HITS algorithm.
Nov 9	Ch 14.3-14.4 Ch 15.1-15.5	PageRank Search ads, Auctions with multiple items, VCG Auctions.
Nov 16	Ch 16.1-16.7	Bayes' Theorem, Herding and information cascades.
Nov 23 **Thanksgiving**	Ch 18.1-18.6	Popularity on the web, Degree distributions, Rich-Get-Richer model, The Music Experiment.
Nov 30	Ch 19.1-19.6	Models of behavior cascades on networks
Dec 7 **Exam 2 Week**	Review Exam 2	**Exam 2 Review** **Exam 2 on December 10, in class**

Student Mental Health and Wellbeing
Students may experience stressors that can impact both their academic experience and personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers the following resources:

- Counseling and Psychological Services (CAPS) - confidential; 734-764-8312; for after-hours urgent support, call and press 0; counseling, workshops, groups and more. Ashley Evearitt, a CAPS counselor is embedded in UMSI, information about how to schedule an appointment with her can be found here.
- Dean of Students Office - 734-764-7420; provides support services to students and manages critical incidents impacting students and the campus community
- Ginsberg Center for Community Service Learning - 734-763-3548; opportunities to engage as learners and leaders to create a better community and world
- Maize and Blue Cupboard (MBC) - 734-936-2794; Food pantry with groceries, kitchen and cooking supplies, personal and household items, and support
- Multi-ethnic Student Affairs (MESA) - 734-763-9044; diversity and social justice through the lens of race and ethnicity
- Office of Student Conflict Resolution - 734-936-6308; offers multiple pathways for resolving conflict
- Office of the Ombuds - 734-763-3545; students can raise questions and concerns about the functioning of the university.
- Services for Students with Disabilities (SSD) - 734-763-3000; accommodations and access to students with disabilities
- Sexual Assault Prevention and Awareness Center (SAPAC) - confidential; 734-764-7771 or 24-hour crisis line 734-936-3333; addresses sexual assault, intimate partner violence, sexual harassment, and stalking
- Spectrum Center - 734-763-4186; support services for LGBTQ+ students
- Trotter Multicultural Center - 734-763-3670; intercultural engagement and inclusive leadership education initiatives
- University Health Service (UHS) - 734-764-8320; clinical services include nurse advice by phone, day or night
- Well-being for U-M Students website - searchable list of many more campus resources
- Wolverine Wellness - 734-763-1320; provides Wellness Coaching, Collegiate Recovery Program, and much more.

Accommodations for Students with Disabilities
The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact

Services for Students with Disabilities https://ssd.umich.edu/; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

Classroom Culture of Care
We are committed to delivering our mission while aiming to protect the health and safety of the community, which includes minimizing the spread of COVID-19. Our entire UM community is responsible for protecting the collective health of all members by being mindful and respectful in carrying out the guidelines laid out in our Wolverine Culture of Care. Individuals seeking to request an accommodation related to the face covering requirement under the Americans with Disabilities Act should contact the Office for Institutional Equity.

In our classrooms all students are expected to adhere to the required safety measures and guidelines of the State of Michigan and the University of Michigan and not come to class when ill or in quarantine.

Lectures
Tuesdays/Thursdays
10:00am – 11:30am
1400 CHEM

Discussions
Discussions will consist of a combination of practice problems and programming exercises related to the material from lectures. Attendance to discussion is mandatory and part of your grade. It is to your benefit to attend in order to work with your peers and GSI on practice exercises. All discussions are in person.

Discussion	GSI	Day/Time	Location
SI 315-002 SI 315-102	Tianliang	Fri 9:00am	2080 SKB
SI 315-003 SI 315-103	Sikandar	Fri 12:00pm	2140 SKB
SI 315-005 SI 315-105	Tianliang	Fri 2:00pm	2200 SKB
SI 315-007 SI 315-107	Sikandar	Fri 3:00pm	1508 NUB

Slack
We will use a Slack workspace for the class. Everyone enrolled in the course should have received an invitation to join the workspace. If you have not received it, please contact us. Participating on Slack is completely optional, and you are welcome to post questions anonymously by beginning your post with /anonymous. We will be monitoring the #assignments and #general channels and trying to answer your questions as promptly as possible. We also encourage you to answer each other's questions and use this platform for discussion related to the class. While we will try to read your posts and clarify or remove wrong or misleading responses, we cannot guarantee that all statements made by students are correct. Please remember to be civil and treat everyone with kindness and respect. Posts (anonymous or otherwise) are expected to adhere to the university's Statement of Student Rights and Responsibilities; posts that fail to meet these standards or that we feel may otherwise create an unsafe learning environment will be removed by the teaching team.

Email
You should feel free to email anyone in the instructional team at any time for any reason. If you would like to email the entire instructional team (instructor, GSIs, IAs) you should send your email to SI315-Fall2026@umich.edu. Emailing the entire team will increase your chances of getting a quick response (compared to emailing a single one of us) since more people will see your email.

Office Hours
We have a total of 10 office hours per week and we are available at a variety of times during the day (see table below). We hope you can find a time that works for you. Feel free to stop by if you have any questions about the class or if you just want to say hello.

Day	Time	Location	Instructor
Monday	10am – 11am	4449 Leinweber And on Zoom	Qingyi
	2pm – 3pm	5136 Leinweber	Daniel
	4pm – 6pm	333 Maynard Room 512	Sikandar
Tuesday	1pm – 3pm	333 Maynard Room 508 And on Zoom	Qingyi
	5pm – 6pm	**TBD**	Sikandar
Wednesday	3pm – 6pm	4460 Leinweber	Tianliang

If you need to meet with the instructor or a GSI and you cannot attend any of the office hours, please contact us to make an appointment.

Assignments
We will have weekly assignments due almost every Wednesday at 11:59pm (with possible exceptions such as breaks, the first week of class, and exams). Working on the

assignments is probably the most beneficial activity for learning the material in this course. They give you an opportunity to engage deeply with the material through a variety of problems of different levels of difficulty. We encourage you to start assignments early after watching all lectures and doing the assigned reading. You should attend office hours, ask questions during discussions, or reach out to us on Slack if you are stuck or confused.

Quizzes
We will have weekly quizzes at the start of each discussion (with possible exceptions such as the first week of class) to help us assess your understanding of key concepts from the course. +r, possibly with small modifications (e.g., new numbers or networks). You will have 10 minutes to answer this question without the use of any aids (e.g., no cheat sheets, calculators, notes, laptops, or internet-connected devices) and with no collaboration from your classmates.

Discussion Worksheets
Discussions provide an opportunity to practice problems and prepare for the week's assignment and future quizzes. Each discussion worksheet will include problems labeled standard, optional, or extra credit. You will work on these problems in preassigned groups, and each group will submit one worksheet at the end of the discussion.

To receive full discussion credit, your group must choose and complete one of the following:

- Make a meaningful attempt at all standard problems, or
- Make a meaningful attempt at all extra-credit problems.

Your solutions do not need to be correct to receive full discussion credit, but your worksheet must demonstrate a substantial effort to attempt every problem in the category your group chooses. You may also work on optional problems for additional practice, but they are not required for credit.

If your group chooses the extra-credit problems, you may earn additional points for each problem solved correctly. Each extra-credit problem will state how many points it is worth. Any extra-credit points earned will be added to the assignment due the following week. Assignment scores are capped at 100%, even after extra credit is applied.

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i-Clicker Questions
Most lectures will have i-Clicker questions. We will use answers to i-Clicker questions to track lecture attendance and participation. Your lecture participation grade will be based on a point system. You collect points by answering questions in the following way:

- 1 point for a **correct answer** if $\geq 50\%$ of students get it right.

- 2 points for a **correct answer** if $<$ 50% of students get it right.
- 1 point for an **incorrect answer** if $<$ 50% of students get it right.

Your lecture participation grade is:

$$\frac{P}{N} + 0.10, \text{ where}$$

P is the number of points you have at the end of the semester.
N is the total number of i-Clicker questions.

Lecture participation grade is capped at 100%
i-Clicker will not be used in discussions.

Grading
Grade breakdown:

Exam 1 (Thu Oct 22 10:00am-11:30am in class): 25%
Exam 2 (Thu Dec 10 10:00am-11:30am in class): 25%
Assignments: 20%
Assignment quizzes: 10%
Discussion section attendance and worksheets: 10%
Lecture attendance: 5%
Lecture i-Clicker questions: 5%

Letter grades:

[98,100] = A+	[87, 90) = B+	[77, 80) = C+	[67, 70) = D+
[93, 98) = A	[83, 87) = B	[73, 77) = C	[63, 67) = D
[90, 93) = A-	[80, 83) = B-	[70, 73) = C-	[60, 63) = D-
			[0, 60) = E

Free Insurance Policy
We understand that unexpected events out of your control, such as illness, job interviews, joining the class late, and family emergencies, happen. That's why we will automatically drop your two lowest grades in each recurring component of the course: assignments, quizzes, discussion sections, and lectures (attendance and participation). No explanations are required to use these "freebies," and our policy is that you should use your drops when you have to miss class, section, assignments, etc. for any reason. By default, we do not allow late submissions. However, if, *after using your free drops*, you face special circumstances (e.g., illness, job interviews, school travel, family emergencies) that require a short extension or lead missed discussion sections, lectures, quizzes, or assignments, you should feel free to contact us as early as possible and, depending on the situation, we will try to work with you to make up the work. **Please do not contact us to request extensions or make-up work before using your two allowed drops**. Documentation and/or involvement of UMSI BSI Academic Advising and Student Success may be necessary.

Extra Credit
From time to time, we may offer the class extra credit opportunities. Extra credit points will be applied to the current week's assignment and scores are capped at 100%. Extra credit points are not meant to increase your grade significantly and only a small number of them will be offered. **Please do not ask for additional extra credit points**.

Collaborating on Assignments
We strongly encourage collaboration while working on some assignments and interpreting readings as a general practice. Active learning is effective. Collaboration with other students in the course will be especially valuable in summarizing the reading materials and picking out the key concepts. You must, however, ***write your assignment submission on your own, in your own words***. You should not allow other students to copy your work or use your solutions to create theirs. If you discussed your assignment with someone before writing it, you must list their names on your written submission.

Use of AI-Generated Text *(this section of the syllabus is adapted from syllabi by Professors Christy Snider and Mike Wakefield, in a handbook compiled by Lance Eaton)*

In the last few years, tools like ChatGPT, Bard, and other open-source tools have made it easy for people to access natural language processing and generative AI technologies. In fact, the University of Michigan recently launched free in-house access to versions of these tools that would normally require payment (https://umgpt.umich.edu). You may find it useful to use the question-answering features of these tools to explore course concepts and better learn the material. In fact, we encourage you to use these tools if they help you deepen your understanding of course content and maybe even proofread and improve the quality of your written explanations. However, please keep the following in mind as you use generative AI:

1. AI is known to regularly fabricate or "hallucinate" seemingly credible information. It can generate wholly inaccurate content that is nonetheless highly persuasive. This is especially true when asking it for references, quotations, citations, and calculations. Additionally, many models have not been trained on, and don't have access to, current data. Don't believe anything that AI tells you. Verify anything it says with a trustworthy source (e.g., textbook, GSI).
2. To get high-quality results from AI, you need to craft good prompts. OpenAI, the company that built ChatGPT, has put out documentation on some of the best practices to use with AI chats to achieve good results.
3. Remember that there are no guarantees or expectations of privacy when using AI, so avoid including any personal information in prompts. Texts and images produced by AI prompts are currently not protected by U.S. copyright law. Additionally, these technologies have large carbon footprints.

As a result:

1. We encourage you not to use AI to solve problems on weekly assignments. AI is not a replacement for knowing and understanding the material, which often happens when you work through problems on your own and allow yourself space to struggle with the material.
2. You will be held accountable for any work you submit. Your grade will be penalized for any mistake or piece of inappropriate text. Therefore, make sure to closely read and verify anything AI generates to ensure that it accurately reflects the facts and your position on a topic or issue.
3. AI is a tool that needs to be cited. If you use AI on an assignment, you need to acknowledge that in a paragraph describing exactly how AI was used and listing the prompts used and output generated. Failure to provide this information is a violation of academic integrity.
4. You must write your assignment submission on your own, in your own words.

Check out Bender et al. (2021), Bender and Koller (2020), and this NY Magazine article for more details about the harms associated with technologies like ChatGPT.

Plagiarism

Unless otherwise specified in an assignment all submitted work must be your own, original work. Any excerpts from the work of others (including the outputs of tools like ChatGPT, as described above) must be clearly identified as a quotation, and a proper citation provided. When collaborating on assignments you are allowed to discuss how to solve problems and explain concepts to each other. However, you are not allowed to copy someone else's solution. Always write solutions independently. Always turn in original work. Always indicate the names of students you collaborate with. All cases of plagiarism will be officially reported.

Any violation of the School's policy on Academic and Professional Integrity will result in serious penalties, which might range from failing an assignment to failing a course, to being expelled from the program. Violations of academic and professional integrity will be reported to UMSI Student Affairs. Consequences impacting assignment or course grades are determined by the faculty instructor; additional sanctions may be imposed by the Senior Associate Dean for Academic Affairs.

Audio and Video Recordings

We will be doing audio and video recordings of all sessions to enable those who cannot attend class in person on a given day to access the content. These recordings will not be made available publicly. Recordings of all sessions will be available on Canvas only to students registered for this class. As part of your participation in this course, you may be recorded. If you do not wish to be recorded, please contact the professor during the first week of class to discuss alternative arrangements. The camera only picks up the front of the room (instructor and slides), but this may require you to sit in a particular place in the room, outside the camera's view. Our classroom has a ceiling mic that picks up student voices. In addition, the instructor's microphone records audio in the room. Students may

not copy and share the lecture videos with those not in the class, or upload them to any other online environment (this is a violation of the Federal Education Rights and Privacy Act (FERPA)).

Students are prohibited from recording/distributing any class activity without written permission from the instructor, except as necessary as part of approved accommodations for students with disabilities. Any approved recordings may only be used for the student's own private use.